Abstract:

Privacy-preserving collaborative learning is increasingly important in sensitive domains such as healthcare, where multiple institutions seek to jointly train models without sharing raw data. Existing distillation-based privacy-preserving frameworks reduce direct exposure to private data by transferring knowledge through distilled samples and teacher outputs. However, they still face three major limitations: distilled data and teacher logits may leak sensitive information, the training pipeline often lacks end-to-end protection, and current methods are not well suited to resolving conflicts among multiple teachers trained on heterogeneous data. To overcome these limitations, we propose a privacy-preserving multi-teacher distillation framework that combines three-party secure computation with knowledge alignment. Specifically, we protect both distilled images and teacher logits via fixed-point encoding and additive secret sharing under a 3PC protocol, preventing the server from directly observing intermediate sensitive information. To address multi-teacher inconsistency, we further introduce a knowledge alignment module with statistical logit normalization, confidence-aware fusion, and label-conditioned correction. Experiments on CIFAR-10 and STL-10 show that the proposed method preserves strong membership privacy while maintaining competitive utility, and yields clear gains in heterogeneous multi-teacher settings. Our results demonstrate that secure computation and adaptive alignment can be jointly integrated into distillation-based learning to achieve both privacy protection and robust collaborative generalization.

1.Introduction:

Deep learning has achieved remarkable success in data-intensive applications such as medical diagnosis, image analysis, and intelligent decision-making. However, many of these applications rely on highly sensitive data, including clinical records, financial information, and personal images, making privacy preservation a central challenge in modern machine learning [1], [2]. In realistic cross-institutional scenarios, multiple organizations often wish to collaboratively train high-quality models, yet direct sharing of raw data is prohibited by privacy regulations and ethical constraints [3]. This tension between collaborative utility and data confidentiality has motivated extensive research on privacy-preserving machine learning [4], [5].

A natural solution is federated learning, which enables multiple clients to jointly train models without sharing raw data [22], [23]. Nevertheless, recent studies have shown that federated learning still suffers from several practical limitations, including high communication overhead, difficulty in handling heterogeneous models or non-IID data, and privacy risks caused by shared model updates or intermediate information [23], [24], [26], [30], [31]. At the same time, modern learning systems are also vulnerable to inference-time privacy attacks. Among them, membership inference attacks (MIAs) are particularly concerning because they aim to determine whether a specific sample was used in model training, thereby revealing potentially sensitive information about individuals [6], [7]. Prior studies have shown that deep neural networks are especially vulnerable to such attacks due to their large capacity and tendency to memorize training

data [8], [9]. Therefore, designing learning frameworks that can simultaneously preserve model utility and reduce privacy leakage has become an important research problem [12], [18], [19].

Existing defenses against MIAs mainly fall into three categories: differential privacy, adversarial or regularization-based methods, and knowledge-transfer-based approaches. Differential privacy provides rigorous theoretical guarantees, and representative frameworks such as PATE protect privacy by aggregating predictions from teacher ensembles with randomized mechanisms [10], [11]. However, these methods often incur substantial utility degradation, especially when the privacy budget is strict or the task becomes more complex [18], [19]. Regularization-based defenses, including perturbed-gradient and confidence-control methods, can improve the privacy-utility trade-off to some extent, but they still rely on modifying the training dynamics of the target model and often provide limited protection under strong attacks [12]–[14]. More recently, knowledge distillation and collaborative distillation have emerged as promising alternatives because they restrict the final student model's direct exposure to private data, reduce communication cost, and naturally support model heterogeneity [15]–[19], [24], [26].

Despite these advantages, existing privacy-preserving distillation frameworks remain insufficient for realistic multi-institution collaboration. First, although distilled data are synthetic, they may still encode statistical properties of the original training set and thus leak private information [19]. Second, teacher outputs, especially logits or soft labels, may expose intermediate knowledge representations when transmitted or processed in plaintext [20], [28]. Third, most existing frameworks are designed around single-teacher settings or implicitly assume consistent teacher behavior, and therefore do not explicitly address the knowledge conflicts caused by heterogeneous training distributions across institutions [25], [27], [29]. These limitations become even more critical in multi-hospital collaboration, where institutions may own data with different resolutions, distributions, and quality levels [30], [31].

To address these challenges, we study privacy-preserving multi-teacher knowledge distillation under heterogeneous data settings. We propose a new framework that integrates a three-party collaborative architecture, pairwise secure channels, and adaptive knowledge alignment for collaborative distillation. Specifically, each institution establishes an independent secure 2-party channel with the server, and both distilled images and teacher logits are protected during transmission through fixed-point encoding and two-share additive secret sharing. This design ensures that no single transmitted share reveals the original content to unauthorized parties. We further introduce a Knowledge Alignment (KA) module to mitigate multi-teacher conflicts via statistical logit normalization, confidence-aware shrinkage, and label-conditioned correction. In our implementation, local teachers are built on a modified ResNet-18 backbone, while the global student adopts a ConvNet architecture. To support heterogeneous cross-dataset experiments, STL-10 samples are resized to 32×32so that they are compatible with CIFAR-10 during collaborative distillation. We organize the overall pipeline into four phases: distilled dataset generation [21], secure teacher inference and sharing, secure reconstruction with knowledge alignment, and final

student distillation training. Experiments on CIFAR-10 and STL-10 show that the proposed method preserves strong membership privacy while maintaining competitive performance. In particular, secure sharing introduces only marginal utility loss, while knowledge alignment provides clear improvements in heterogeneous multi-teacher settings. In summary, this work makes three main contributions: 1) extending privacy-preserving distillation from single-teacher to multi-teacher collaborative settings with a three-party secure training architecture based on pairwise 2PC channels; 2) proposing a knowledge alignment strategy to explicitly address teacher inconsistency under heterogeneous data distributions; and 3) demonstrating through extensive experiments a favorable balance among privacy, utility, and cross-domain collaborative generalization.

2.Related Work:

A. Membership Inference Attacks and Privacy Defenses

Membership inference attacks (MIAs) have become one of the most widely studied privacy threats in modern machine learning. Early work showed that an adversary can infer whether a target sample belongs to the training set of a model by exploiting prediction confidence and output patterns [6]. Subsequent studies further demonstrated that stronger white-box attackers can additionally leverage losses, gradients, and internal model states, making deep neural networks particularly vulnerable in both centralized and federated settings [7]. This vulnerability is closely related to memorization and overfitting, as large-capacity models tend to retain training-specific information that can later be exploited by privacy attacks [8], [9].

To mitigate these risks, existing defenses can be broadly grouped into differential-privacy-based methods, regularization-based methods, and knowledge-transfer-based methods. Differential privacy provides formal privacy guarantees by perturbing the learning process, with DP-SGD and PATE being representative examples [10], [11]. However, such methods often suffer from a significant privacy–utility trade-off, especially under strict privacy budgets or more challenging tasks [18], [19]. Regularization-based defenses reduce leakage by suppressing overfitting, for example through adversarial regularization, perturbed-gradient optimization, or confidence-control strategies [12]–[14]. Although these methods can improve the empirical privacy–utility balance, they generally modify the target model's optimization process and may still offer limited robustness under strong attacks [12], [13]. In contrast, knowledge-transfer-based approaches aim to decouple the final model from direct access to private data, providing a different route to privacy preservation [15], [16], [18], [19].

B. Knowledge Distillation and Privacy-Preserving Distillation

Knowledge distillation was originally proposed as a model compression technique, where a compact student model learns from the softened outputs of a teacher model [15]. Since then, it has evolved into a broad research area, with surveys summarizing its variants in teacher–student transfer, mutual learning, and self-distillation [16], [17]. In privacy-sensitive settings, this paradigm is particularly attractive because the student model can be trained on transferred knowledge rather than on raw private data. DMP shows that membership privacy can be improved by carefully selecting reference data and transferring low-entropy teacher knowledge to the student [18]. PLDK further combines dataset distillation with knowledge distillation, using a compact distilled dataset and teacher logits to train a student model while reducing direct exposure to private data [19].

Despite these advantages, most privacy-preserving distillation methods still assume relatively simple teacher–student settings. In many cases, they rely on a single teacher or assume that transferred teacher knowledge is sufficiently reliable and directly usable by the student [18], [19]. Such assumptions become less realistic in collaborative environments where different institutions may own data from different distributions and train teachers with varying quality. Under these conditions, inconsistent soft labels, biased logits, and heterogeneous local knowledge can all degrade the effectiveness of student supervision. Therefore, while privacy-preserving distillation provides a promising alternative to direct private-data training, its robustness under heterogeneous multi-teacher settings remains insufficiently addressed.

C. Dataset Distillation and Distilled Data as Knowledge Carriers

Dataset distillation aims to compress a large dataset into a much smaller synthetic set that preserves the original training utility [20], [21]. Early work formulated dataset distillation as optimizing synthetic examples that can reproduce model performance after training [20]. More recent methods, such as matching training trajectories, significantly improved distilled-data quality by aligning the optimization behavior induced by synthetic and real datasets [21]. These advances make dataset distillation especially relevant to privacy-preserving learning, since distilled data can serve as a compact carrier of transferable knowledge.

However, using distilled data in privacy-sensitive scenarios also introduces new concerns. Although synthetic samples are not raw private records, they are still optimized from the private training distribution and may encode task-relevant statistical properties of the source data [19]. This means that distilled data can reduce direct data exposure but do not automatically eliminate privacy risks. Accordingly, methods that rely on distilled samples for collaborative learning must consider not only how to generate informative distilled data, but also how to securely handle and transmit them during subsequent training and inference stages.

D. Federated Learning and Federated Distillation

Federated learning (FL) enables multiple clients to jointly train models without sharing raw data and has become a major framework for privacy-aware distributed learning [22], [23]. Nevertheless, standard FL still faces several practical challenges, including communication overhead, vulnerability to leakage through gradients or model updates, and poor adaptation to non-IID data and heterogeneous local models [23], [30], [31]. These issues are especially pronounced in real-world scenarios such as healthcare networks, where institutions often differ substantially in data distributions, computational resources, and model capacities [3], [30]. In heterogeneous cross-dataset settings, additional preprocessing is often required to unify incompatible input spaces, which is also relevant to our CIFAR-10/STL-10 setting.

To address these limitations, federated distillation (FD) has been proposed as an alternative collaborative paradigm that exchanges predictions or logits instead of full model parameters. FedKD improves communication efficiency by performing knowledge distillation between compact shared models and larger local models [24]. FedMD further demonstrates that model distillation can support heterogeneous local architectures, making collaborative learning more flexible than standard parameter averaging [26]. More recently, Selective-FD shows that collaborative distillation is not always reliable in heterogeneous settings: when there is no sufficiently strong teacher, local knowledge can become misleading or ambiguous, and indiscriminate sharing may even harm the final performance [25]. These findings indicate that collaborative distillation must consider not only efficiency, but also the reliability and compatibility of exchanged knowledge.

E. Personalized and Multi-Teacher Collaborative Distillation

Another related line of work studies personalized or multi-source knowledge transfer in distributed systems. Deep mutual learning shows that multiple models can improve one another through mutual supervision, even without a fixed teacher–student hierarchy [17]. Learning from multiple teacher networks further demonstrates the value of aggregating knowledge from several teachers rather than relying on a single supervisory source [29]. In federated settings, personalized mutual distillation methods such as FedMKD use bidirectional knowledge transfer and regularization to address client-specific objectives and alleviate degradation caused by heterogeneous local data [27]. Recent work has also explored privacy-preserving knowledge distillation for latency-critical federated task offloading, highlighting the practical need to protect transferred knowledge under communication and deployment constraints [28].

Although these studies show that multi-model and personalized distillation can improve flexibility and performance, their focus is typically on personalization, communication efficiency, or task-level optimization rather than end-to-end privacy protection of exchanged intermediate knowledge. In particular, existing methods usually do not simultaneously address two coupled challenges that are central in our setting: securing distilled samples and teacher logits during collaborative training, and explicitly aligning heterogeneous multi-teacher knowledge before student distillation. This gap is particularly important in cross-institution scenarios, where privacy protection must extend beyond raw data to all exchanged supervisory signals.

F. Secure Multi-Party Computation for Privacy-Preserving Learning

Beyond distillation and federated optimization, secure multi-party computation (MPC) provides another important line of research for privacy-preserving machine learning. In general, MPC-based systems aim to perform data processing or model-related computation without revealing the underlying private inputs to other participants. In practice, software frameworks such as NssMPClib provide reusable primitives for secret sharing, arithmetic operations, and collaborative protocol construction in MPC-based systems [32]. Such tools make it easier to build secure transmission and reconstruction pipelines over protected tensors.

Our work is closely related to this direction, but differs in an important way. We do not use secure computation as a standalone full encrypted training framework. Instead, we use it as a mechanism to protect the transmission of distilled samples and teacher logits within a collaborative distillation pipeline. More specifically, the secure layer in our prototype is realized through pairwise secure channels in a three-party architecture, together with fixed-point encoding and additive secret sharing, rather than end-to-end encrypted optimization over the entire model training process.

G. Summary

In summary, prior work has made substantial progress in MIA defense, knowledge distillation, dataset distillation, federated learning, collaborative distillation, and secure computation [6]–[32]. Differential privacy and regularization provide important baselines for privacy protection, but they often face limitations in utility or robustness [10]–[14]. Privacy-preserving distillation improves this trade-off by reducing the student model's direct access to private data [18], [19], while federated and collaborative distillation improve communication efficiency and support heterogeneous local architectures [24]–[26]. Secure computation, on the other hand, offers a practical way to protect intermediate values during collaborative learning [32]. However, a clear gap remains for realistic multi-institution settings where distilled samples and teacher logits themselves require protection, and where heterogeneous teachers may provide conflicting knowledge. Our work is positioned at

this intersection by jointly considering secure protection of exchanged distilled knowledge and explicit alignment of multi-teacher supervision under heterogeneous data distributions.

3.Our Methodology:

3.1 Overview

We propose a privacy-preserving multi-teacher knowledge distillation framework for heterogeneous cross-institution collaboration. As illustrated in **Fig. 1**, the overall system consists of two client institutions and a central server. The full collaborative pipeline includes four sequential phases: local preparation, secure data sharing, secure reconstruction and knowledge alignment, and student distillation training. The technical details of the representative client-side pipeline and the server-side pipeline are further illustrated in **Fig. 2(a)** and **Fig. 2(b)**, respectively. The goal is to collaboratively train a student model without exposing either raw private data or intermediate teacher knowledge throughout the distillation pipeline.

Different from existing single-teacher or plaintext distillation methods, our framework addresses two coupled challenges. First, the distilled samples and teacher logits used for collaborative training may still reveal sensitive information if directly transmitted or processed. Second, teacher models trained on heterogeneous institutional data often produce inconsistent predictions, and naive aggregation may introduce noisy or conflicting supervision. To address these issues, we combine a three-party collaborative architecture with secure secret sharing and an adaptive Knowledge Alignment (KA) module. In our implementation, the secure communication layer is built on NssMPClib [32], and the distilled data generation stage follows the matching training trajectories (MTT) strategy [21].

It is important to clarify that our system is a three-party collaborative framework, but the secure communication primitives are instantiated through independent 2PC channels between each hospital and the server. Therefore, the framework should be understood as a three-party secure collaborative training pipeline with pairwise secure sharing, rather than a fully joint encrypted training procedure over a monolithic 3PC arithmetic protocol.

3.2 Problem Formulation

Assume there are M institutions, and the m-th institution owns a private dataset

$$\mathcal{D}_m = \{(x_i^{(m)}, y_i^{(m)})\}_{i=1}^{N_m},$$

where $x_i^{(m)} \in \mathbb{R}^{H\times W\times C}$ denotes an input sample and $y_i^{(m)} \in \{1, \dots, K\}$ is its corresponding label. Due to privacy regulations and institutional constraints, the raw datasets $\{\mathcal{D}_m\}_{m=1}^{M}$ cannot be directly shared.

Our objective is to train a global student model S_θ by distilling knowledge from multiple local teacher models $\{T_m\}_{m=1}^{M}$, while satisfying the following requirements:

1. Raw-data privacy: no institution reveals its original training data to any other party.
2. Intermediate-knowledge privacy: distilled samples and teacher logits should be protected during transmission and collaborative training.
3. Heterogeneous knowledge fusion: the student should effectively learn from multiple teachers even when their output distributions differ significantly.

To achieve this goal, we design a four-phase framework that combines dataset distillation, secure sharing, knowledge alignment, and student distillation training.

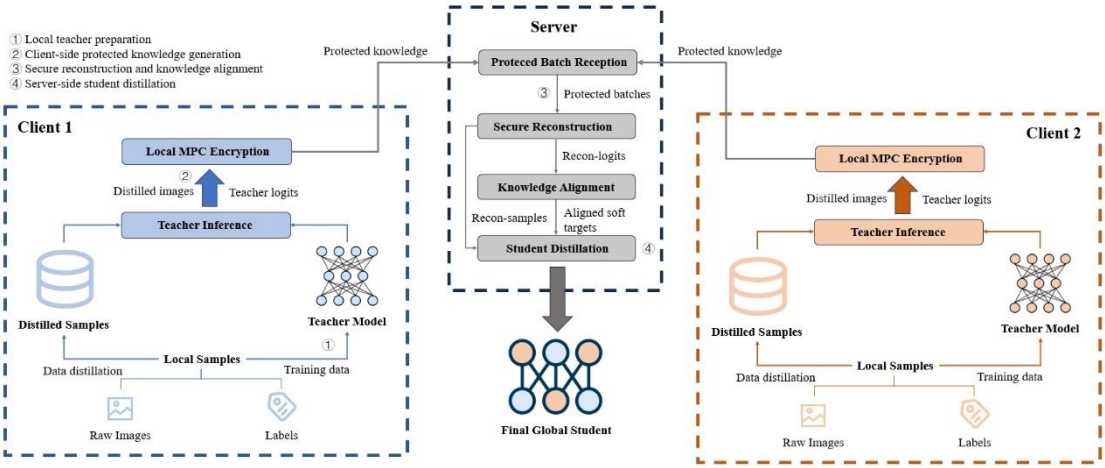

Fig. 1. Overall framework of the proposed secure collaborative distillation pipeline.

3.3 Phase I: Local Preparation

3.3.1 Distilled Data Generation

In the first phase, each institution prepares a compact distilled dataset from its local data. Let

$$\mathcal{S}_m = \{(s_j^{(m)}, \tilde{y}_j^{(m)})\}_{j=1}^{K_m},$$

where $K_m \ll N_m$. Following dataset distillation by matching training trajectories [21], the synthetic samples are optimized such that training a model on $\mathcal{S}_m$ induces a parameter trajectory similar to that produced by the real local data. A representative trajectory-matching objective is

$$\mathcal{L}_{\text{traj}} = \frac{\| \hat{\theta}_{t+N} - \theta^*_{t+M} \|_2^2}{\| \theta^*_t - \theta^*_{t+M} \|_2^2}. \qquad (1)$$

In the full framework, each institution can locally generate distilled samples via MTT. In the current implementation, pre-generated distilled datasets are loaded from disk and used as compact training carriers in later secure collaborative distillation. The qualitative visualization of these distilled samples is reported later in **Fig. 3** in the experiments section.

3.3.2 Local Teacher Training

Each institution also trains a local teacher model T_m on its private dataset $\mathcal{D}_m$. During inference, the teacher consumes an augmented view of the distilled sample:

$$z^{(m)} = T_m(\text{Aug}(s)). \qquad (2)$$

This is important because the teacher logits are generated from augmented inputs, whereas the protected image tensor transmitted to the server corresponds to the original distilled image. This decouples teacher-side and student-side augmentations. This local preparation process is explicitly illustrated in the representative client-side pipeline shown in Fig. 2(a).

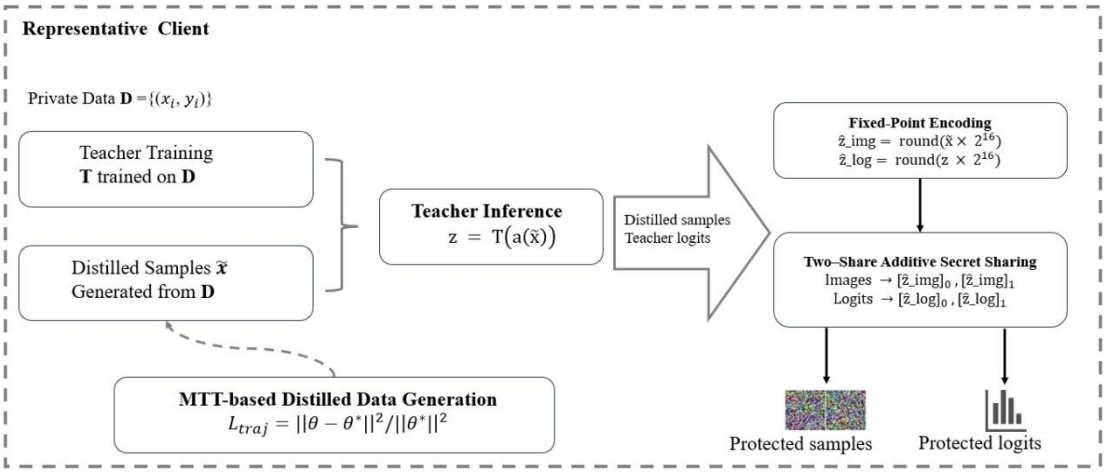

Fig. 2(a) Technical details of the representative client-side pipeline.

3.4 Phase II: Secure Data Sharing via Pairwise 2PC Channels

After local preparation, each institution performs teacher inference on its distilled samples and securely transmits both distilled images and teacher logits. The client-side fixed-point encoding and two-share additive secret sharing procedure are illustrated in Fig. 2(a).Directly sharing these tensors in plaintext would expose intermediate knowledge to the server. Therefore, before transmission, all floating-point values are encoded into fixed-point ring representations:

$$\hat{x} = \mathrm{round}(x \cdot 2^{f}), \qquad (3)$$

where f is the scaling factor. In our implementation, $f = 16$.

The encoded tensors are then protected through additive secret sharing. In particular, this procedure is applied to both the distilled sample tensor and the teacher-logit tensor, producing protected sample shares and protected logit shares, as illustrated in Fig. 2(a). Since the implementation uses a 64-bit integer ring, we have

$$\hat{x} = [\hat{x}]_0 + [\hat{x}]_1 (\mathrm{mod} 2^{64}), \qquad (4)$$

where $[\hat{x}]_0$ and $[\hat{x}]_1$ are two random additive shares. These shares are distributed through secure communication channels, while the overall system operates in a three-party setting involving Hospital A, Hospital B, and the server. Concretely, each hospital establishes an independent secure 2PC channel with the server using the primitives provided by NssMPClib [32]. Thus, the framework is a three-party collaborative system with pairwise secure channels, rather than a monolithic 3-share protocol.

This secure encoding-and-sharing process is applied to both:

- the distilled image tensor $\hat{z}_{\mathrm{img}}$, and
- the teacher logits tensor $\hat{z}_{\mathrm{log}}$.

By construction, the local view of each share is visually random and does not reveal semantic content. The corresponding share visualizations are shown later in **Fig. 4** in the experiments section. Finally, the class labels are transmitted in plaintext. This is consistent with our threat model: labels are integer indices and do not themselves reveal visual sample features. Therefore, protecting distilled images and teacher logits is the central privacy requirement in our setting.

3.5 Phase III: Secure Reconstruction and Knowledge Alignment

3.5.1 Secure Reconstruction

After transmission, the protected tensor is reconstructed and decoded within the secure online workflow as

$$x_{\mathrm{rec}} = \frac{([\hat{x}]_0 + [\hat{x}]_1 mod 2^{64})}{2^f}. \qquad (5)$$

In practice, this reconstruction procedure is applied to both protected sample shares and protected logit shares, yielding reconstructed distilled samples and reconstructed teacher logits, respectively, as illustrated in **Fig. 2(b)**.This means that the server does not receive plaintext distilled images or logits directly from the institutions; instead, it reconstructs protected tensors inside the collaborative training pipeline. The server-side reconstruction process is illustrated in **Fig. 2(b)**, while the qualitative reconstruction fidelity is shown later in **Fig. 5** in the experiments section.

3.5.2 Offline Statistical Fitting

Before student training starts, the server performs a one-time offline fitting step over the received batches from all institutions. This step computes the source-wise and global statistics required by Knowledge Alignment, including the source mean and standard deviation $(\mu_{src}, \sigma_{src})$, the global mean and standard deviation (μ_g, σ_g), and the confidence prior w. These statistics are computed once before training, and all later batch-wise alignment operations only involve lightweight tensor transformations, without introducing additional communication rounds.

3.5.3 Knowledge Alignment

Although secure reconstruction protects privacy, it does not resolve the inconsistency among teachers trained on heterogeneous data. In our setting, the KA module operates on batches that originate from a single source teacher at a time. Therefore, the alignment step does not perform real-time multi-teacher weighted summation over the same sample. Instead, it aligns the logits of the current source teacher by shrinking them toward a global canonical distribution.

Let $z^{(m)} \in \mathbb{R}^K$denote the logits of teacher m, and let

$$\boldsymbol{\mu}_m, \boldsymbol{\sigma}_m, \boldsymbol{\mu}_g, \boldsymbol{\sigma}_g \in \mathbb{R}^K$$

be the **class-wise** mean and standard deviation vectors of the local teacher and the global teacher distribution, respectively. The statistical alignment step is defined as

$$\tilde{z}^{(m)} = \frac{z^{(m)} - \boldsymbol{\mu}_m}{\boldsymbol{\sigma}_m} \odot \boldsymbol{\sigma}_g + \boldsymbol{\mu}_g, \qquad (6)$$

where $\odot$denotes element-wise multiplication.

Next, we compute a dataset-level confidence prior for each teacher. Let $\mathcal{S}_m$denote the set of distilled samples associated with teacher m. The class-wise mean confidence is

$$\bar{p}_c^{(m)} = \frac{1}{|\mathcal{S}_m|} \sum_j \left[\mathrm{softmax}\left(\frac{z_j^{(m)}}{\tau} \right) \right]_c, \qquad (7)$$

where $\tau$is the temperature parameter. The corresponding confidence weight is

$$w_c^{(m)} = \frac{\bar{p}_c^{(m)}}{\sum_{r=1}^{M} \bar{p}_c^{(r)} + \epsilon}, \qquad (8)$$

where $\epsilon$is a small constant. Note that $w_c^{(m)}$is a pre-computed dataset-level statistic, not a per-sample quantity.

For a batch originating from teacher m, the aligned logits are obtained by shrinking the statistically aligned logits toward the global mean:

$$z^{\text{blend}} = \mathbf{w}^{(m)} \odot \tilde{z}^{(m)} + \left(1 - \mathbf{w}^{(m)}\right) \odot \boldsymbol{\mu}_g, \qquad (9)$$

where $\mathbf{w}^{(m)} = [w_1^{(m)}, \dots, w_K^{(m)}] \in \mathbb{R}^K$. This is different from directly summing multiple teacher logits over the same sample. Instead, each batch is aligned according to its own source teacher and then enters a unified student training stream.

Finally, to avoid semantic flipping between soft supervision and hard labels, we apply a label-conditioned correction:

$$z_c^{\text{align}} = \begin{cases} z_c^{\text{blend}} + \lambda\, \sigma_{g,c}, & c = y, \\ z_c^{\text{blend}}, & c \neq y, \end{cases} \lambda = 0.5, \qquad (10)$$

where $\sigma_{g,c}$ denotes the c-th element of the global standard deviation vector.

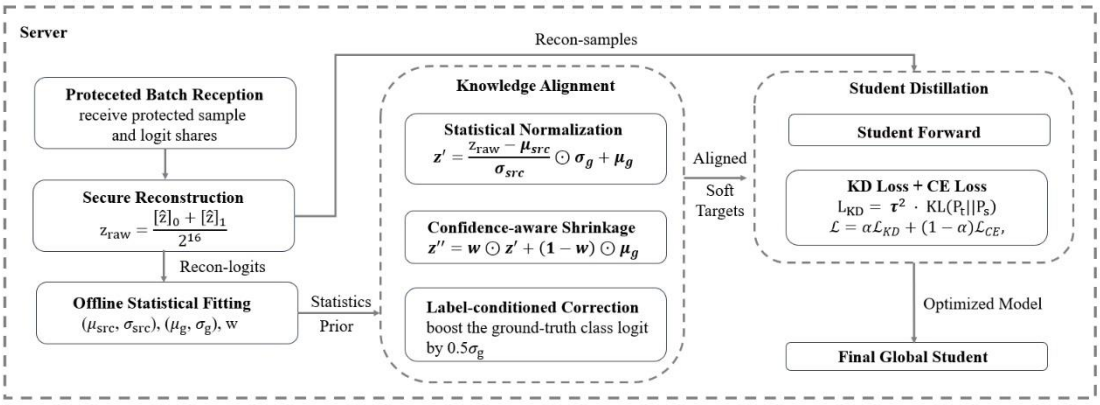

Fig. 2(b) Technical details of the server-side reconstruction, alignment, and distillation pipeline.

3.6 Phase IV: Student Distillation Training

The final student model S_θ is trained on the reconstructed distilled data using both hard labels and aligned teacher supervision. Let z_S denote the student logits and z^{align} denote the aligned teacher logits. As illustrated in Fig. 2(b), reconstructed samples are fed into the student forward path, while the aligned teacher logits are used to form the soft supervisory signal in the distillation loss.We use the standard distillation objective

$$\mathcal{L}_{KD} = \tau^2\, \text{KL}\left(\text{softmax}\left(\frac{z^{\text{align}}}{\tau}\right) \;\middle\|\; \text{softmax}\left(\frac{z_S}{\tau}\right) \right), \qquad (11)$$

where the teacher distribution serves as the target distribution.

We further use the hard-label cross-entropy loss

$$\mathcal{L}_{CE} = -\sum_{c=1}^{K} y_c \log p_S(c), \qquad (12)$$

where $p_S(c)$ is the student probability for class c. The final objective is

$$\mathcal{L} = \alpha \mathcal{L}_{KD} + (1 - \alpha)\mathcal{L}_{CE}, \qquad (13)$$

where $\alpha \in [0,1]$ balances soft and hard supervision.

In implementation, the reconstructed batches from the two institutions are first merged into a single

training pool and then randomly shuffled at every epoch. This design ensures that samples from different institutions are interleaved during student training, which improves optimization stability and reduces the risk of sequential bias. In addition, the reconstructed images are independently augmented again before student forward propagation. As a result, the teacher and the student observe two different augmented views of the same distilled sample, which acts as a consistency-enhancing dual-view training strategy.

3.7 Discussion

Compared with existing privacy-preserving distillation methods, the proposed framework differs in two essential aspects. First, it extends privacy protection from raw data to the entire collaborative distillation pipeline by protecting both distilled images and teacher logits through secure secret sharing over pairwise 2PC channels in a three-party architecture. Second, it does not treat multi-teacher supervision as a naive real-time fusion problem; instead, it aligns each source teacher distribution toward a shared global canonical space using offline statistics and confidence-aware shrinkage. Therefore, the proposed methodology unifies privacy protection and heterogeneous knowledge stabilization in a single collaborative distillation framework.

4. Experiments

4.1 Experimental Setup

4.1.1 Datasets

We evaluate the proposed framework on two benchmark image classification datasets, CIFAR-10 and STL-10. CIFAR-10 contains 50,000 training images and 10,000 test images with a native resolution of 32×32, while STL-10 contains 5,000 training images and 8,000 test images with a native resolution of 96×96. Both datasets contain 10 semantic classes. To support heterogeneous cross-dataset collaboration, all STL-10 samples are resized to 32×32, making them compatible with CIFAR-10 in the unified collaborative distillation pipeline. This preprocessing step is necessary for both homogeneous STL-10 experiments and cross-domain CIFAR-10/STL-10 collaboration.

4.1.2 Distilled Data Generation

Following the trajectory matching strategy in dataset distillation [21], we generate distilled datasets with 50 images per class (IPC = 50) for both CIFAR-10 and STL-10. For STL-10, all images are first resized to $\mathbf{32 \times 32}$to ensure compatibility with the student model and to enable cross-domain collaborative experiments. The distilled data generation is performed independently by each institution on its private dataset, and the resulting synthetic samples are used as compact knowledge carriers in the subsequent secure distillation pipeline. This design reduces direct exposure to raw private data while retaining task-relevant supervision for collaborative training. The generated distilled samples are visualized in Fig. 3, where they remain class-discriminative despite being synthetic and highly compressed.

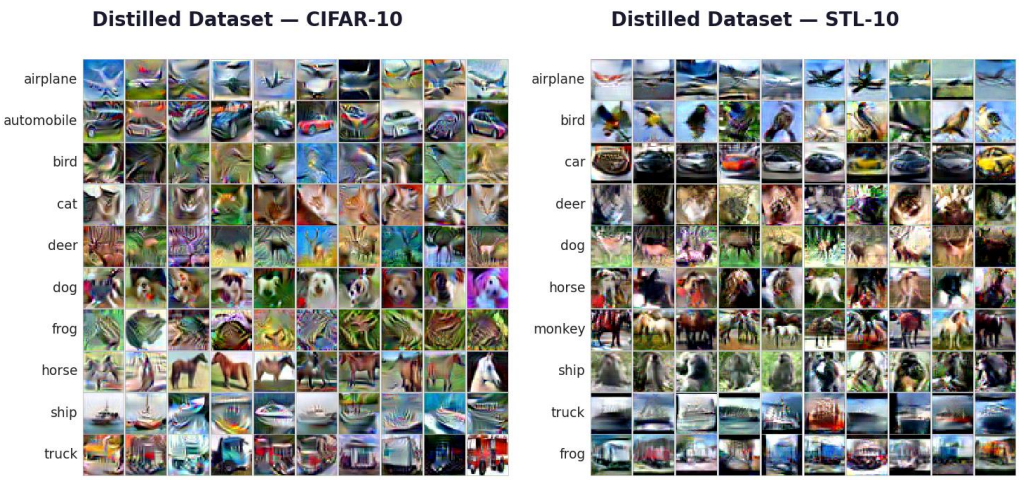

Fig. 3. (a) Distilled data for CIFAR-10 , (b) Distilled data for STL-10

4.1.3 Model Architecture and Teacher Configuration

Local teacher models are implemented using a modified ResNet-18 architecture, whereas the distilled student model in our framework adopts a lightweight ConvNet. This teacher–student asymmetry is consistent with the objective of compressing stronger private models into an efficient collaborative student. For the standard defense baselines No Defense, Regu (WD+LS), and DMP, we report results under the AlexNet setting. This choice follows the conventional defense-oriented evaluation protocol and provides a stable reference point for comparison, while our distillation-based framework is evaluated under its own teacher–student setting.

On CIFAR-10, Teacher A trained on the full training set achieves 95.36% accuracy and Teacher B trained on a subset achieves 80.61% accuracy. On STL-10, Teacher A trained on the full training set achieves 84.95% accuracy and Teacher B trained on a subset achieves 71.08% accuracy. These teacher pairs simulate heterogeneous institutional supervision with different data quality levels and are used to construct both homogeneous and heterogeneous collaborative settings.

Table 1. Dataset statistics and teacher configurations.

Dataset	Train Size	Test Size	Resolution	Teacher A Acc.	Teacher B Acc.
CIFAR-10	50,000	10,000	32×32	95.36%	80.61%
STL-10	5,000	8,000	96×96 → 32×32	84.95%	71.08%

4.1.4 Evaluation Metrics

We report both utility and privacy metrics. Utility is measured by test accuracy (Test Acc) on the held-out test set.

To quantify privacy leakage, we report Privacy Accuracy (PrivAcc) under membership inference attacks. PrivAcc measures the attack success rate in distinguishing member samples from non-member samples. A higher PrivAcc indicates that the target model leaks more membership information, whereas values close to 50% indicate that the attacker performs no better than random guessing. Therefore, from a privacy perspective, the goal is to keep PrivAcc as close as possible to 50%.

We consider both black-box attacks $(\boldsymbol{A_b})$, which only access prediction confidence vectors [6], and white-box attacks $(\boldsymbol{A_w})$, which additionally exploit richer model-side information such as gradients [7]. The corresponding privacy metrics are denoted by Priv Acc (Ab) and Priv Acc (Aw), respectively. In Table 2 and Table 3 we report both Priv Acc (Ab) and Priv Acc (Aw). For the

remaining tables (Tables 4-7), we report only Priv Acc (Ab) under the unified black-box MIA protocol for clarity and consistency.Following the strong-adversary setting adopted in prior membership privacy studies [18], [19], the attack models are trained using 12,500 member samples and 12,500 non-member samples from each dataset. Under this protocol, values significantly above 50% indicate substantial privacy leakage, while values slightly below 50% suggest that the attacker is effectively misled.

4.2 Main Results and Privacy–Utility Comparison

We first compare the proposed framework with representative baselines on both CIFAR-10 and STL-10. The unified results are summarized in Table 2, which reports the main privacy–utility comparison of standard defenses, distillation-based baselines, and our secure collaborative variants across the two datasets. Similar to prior privacy-preserving distillation studies, this table is intended to provide a consolidated view of utility and membership privacy under a common evaluation protocol, rather than to restrict all methods to an identical training pipeline. To ensure interpretability across heterogeneous protocols, we distinguish between in-setting comparison, cross-setting comparison, and cross-paradigm comparison throughout the experimental section. Specifically, Table 2 serves as the main privacy–utility comparison, Table 3 compares against heterogeneous distillation baselines under their native collaborative protocols, and Table 4 compares against representative methods from differential privacy and regularization paradigms.

Several observations can be drawn from these results. First, No Defense consistently yields the most severe privacy leakage. On CIFAR-10, the attack accuracies reach 76.80% and 77.20% under black-box and white-box attacks, respectively. On STL-10, the corresponding values remain similarly high at 73.18% and 78.20%. These results confirm that high predictive performance alone does not imply privacy preservation.

Second, regularization-based defense reduces privacy leakage on CIFAR-10, but remains insufficient on the more challenging STL-10 benchmark. In particular, Regu (WD+LS) reduces privacy risk on CIFAR-10 to the low-50% range, but on STL-10 the attack accuracies remain around 68%, despite achieving 55.73% test accuracy. This suggests that regularization alone is not robust enough to suppress membership leakage under more difficult visual settings.

Third, distillation-based methods provide a more favorable privacy–utility trade-off. DMP reduces attack accuracy to nearly random guessing on both datasets while maintaining moderate utility. PLDK further improves utility on CIFAR-10 and preserves privacy safety on STL-10, indicating that knowledge transfer through distilled data is a strong alternative to direct training-based defenses.

Finally, our secure variants show that privacy protection can be integrated with collaborative distillation while remaining competitive in utility. On CIFAR-10, PLDK + MPC achieves 65.61% accuracy with privacy metrics close to 50%. In the multi-teacher setting, MPC + KA reaches 64.36%, with black-box and white-box privacy accuracies of 49.62% and 49.94%, respectively. On STL-10, the benefit of alignment is even more pronounced: PLDK + MPC reaches 41.88%, whereas MPC + KA increases the accuracy to 49.51%, corresponding to a 7.63% gain while keeping privacy attack accuracy around random guessing. These results demonstrate that the proposed framework preserves strong membership privacy on both datasets and that KA becomes increasingly important as the task becomes more challenging.

Table 2. Privacy–utility trade-off comparison (Accuracy %).

Dataset	Method	Model	Teacher Setting	Test Acc (%)	Priv Acc (Ab) ↓	Priv Acc (Aw) ↓
CIFAR-10	No Defense†	AlexNet	Single	67.46	76.80	77.20
	Regu (WD+LS)†	AlexNet	Single	53.20	53.00	53.80
	DMP†	AlexNet	Single	65.00	50.60	51.30
	PLDK	ConvNet	Single	69.30	50.21	50.28
	PLDK + MPC	ConvNet	Single (95%)	65.61	50.22	50.29
	MPC + KA (Ours)	ConvNet	95% + 80%	64.36	49.62	49.94
STL-10	No Defense†	AlexNet	Single	59.58	73.18	78.20
	Regu (WD+LS)†	AlexNet	Single	55.73	68.02	67.99
	DMP†	AlexNet	Single	45.86	51.08	51.31
	PLDK	ConvNet	Single (85%)	40.90	49.38	49.32
	PLDK + MPC	ConvNet	Single (85%)	41.88	49.25	49.75
	MPC + KA (Ours)	ConvNet	85% + 71%	49.51	49.85	49.92

† denotes baseline methods outside our secure multi-teacher framework.

4.3 Cross-setting Comparison with Heterogeneous Distillation Baselines

To better position the proposed framework against representative heterogeneous collaborative distillation methods, we further compare it with FedMD, FedKD, and Selective-FD. The corresponding results are reported in **Table 3**. This comparison should be interpreted as a **cross-setting comparison**, rather than a strictly protocol-matched one. In particular, FedMD, FedKD, and Selective-FD are originally designed under native full local-data collaborative distillation protocols, where local institutions directly participate in collaborative training using their full training data. In contrast, our method uses a distilled-data-driven secure collaboration protocol: each institution first compresses its private knowledge into distilled samples, then protects the distilled samples and teacher logits through secure sharing, and finally performs student distillation on reconstructed knowledge at the server side. Therefore, the purpose of Table 3 is not to claim identical training conditions across all methods, but to compare their overall privacy–utility behavior under heterogeneous collaborative learning.

Another important point is that the original papers of FedMD, FedKD, and Selective-FD do not report the membership privacy metric adopted in this work. Their published results mainly focus on utility under their own protocols, but do not provide directly comparable privacy measurements in our threat model. To make the comparison fair and interpretable, we re-evaluate all methods in Table

3 under the same membership inference attack protocol described in Section 4.1.4, and report the corresponding black-box and white-box privacy accuracies as Priv Acc (Ab) and Priv Acc (Aw), respectively. In this sense, the test accuracy values reflect each method's performance under its own collaborative setting, while the privacy values are re-measured under a unified MIA evaluation standard. This unified re-evaluation is essential, because otherwise the privacy behavior of heterogeneous distillation baselines would remain incomparable.

Several observations can be made from Table 3. First, FedMD and FedKD achieve substantially higher utility on CIFAR-10, with test accuracies of 79.37% and 78.66%, respectively, but both exhibit severe privacy leakage. Their Priv Acc (Ab) and Priv Acc (Aw) are far above random guessing, showing that heterogeneous collaborative distillation without protected knowledge transmission may preserve utility at the cost of considerable membership exposure. This result is consistent with the intuition that direct exchange of distillation knowledge, even if effective for accuracy, can still leave strong attack signals for membership inference.

Second, Selective-FD serves as a stronger privacy-aware heterogeneous distillation baseline. On CIFAR-10, it reaches 68.98% accuracy, which is higher than our 64.36%. However, its privacy leakage remains noticeably larger than ours, with Priv Acc (Ab) = 51.18% and Priv Acc (Aw) = 54.44%, compared with 49.62% and 49.94% for our method. This result is important because it shows that even a selection-enhanced heterogeneous distillation strategy does not fully eliminate privacy leakage when the exchanged knowledge itself is not protected. In contrast, our framework explicitly protects both distilled samples and teacher logits during transmission, which allows the attack accuracy to remain much closer to random guessing.

Third, on STL-10, the proposed method achieves the best utility among all compared heterogeneous distillation baselines, reaching 48.60% accuracy. FedMD and FedKD remain below 44%, while Selective-FD further drops to 34.89%. At the same time, our privacy accuracies remain near random guessing, with Priv Acc (Ab) = 50.05% and Priv Acc (Aw) = 49.92%. This suggests that the proposed framework is not only privacy-preserving, but also more robust under the more challenging STL-10 setting. In other words, when the task becomes harder and the heterogeneous collaboration problem becomes more unstable, secure sharing and knowledge alignment appear to offer greater advantages than direct heterogeneous distillation.

Overall, Table 3 shows a clear trade-off pattern. Methods such as FedMD and FedKD may achieve stronger utility under their native full-data collaborative settings, but they also suffer from pronounced privacy leakage. Selective-FD provides a stronger utility–privacy balance than conventional heterogeneous distillation methods, yet it still cannot match the privacy stability of our framework. By protecting distilled samples and teacher logits and by aligning heterogeneous teacher supervision after secure reconstruction, the proposed method achieves a more favorable privacy-preserving trade-off, especially when privacy control is regarded as a first-class objective rather than a secondary constraint.

Table 3. Cross-setting comparison with heterogeneous distillation baselines.

Dataset	Method	Setting	Test Acc (%)	Priv Acc (Ab) ↓	Priv Acc (Aw) ↓
CIFAR-10	FedMD	Heterogeneous FD	79.37	69.25	72.91
	FedKD	Collaborative KD	78.66	73.88	79.17
	Selective-FD	Heterogeneous FD +	68.98	51.18	54.44

Dataset	Method	Setting	Test Acc (%)	Priv Acc (Ab) ↓	Priv Acc (Aw) ↓
		selection			
	Ours(distilled 500 images)	Secure distillation + MPC	64.36	49.62	49.94
STL-10	FedMD	Heterogeneous FD	43.52	49.85	50.55
	FedKD	Collaborative KD	43.09	49.85	50.42
	Selective-FD	Heterogeneous FD + selection	34.89	49.95	51.08
	Ours(distilled 500 images)	Secure distillation + MPC	48.60	50.05	49.92

4.4 Cross-paradigm comparison with differential privacy and regularization baselines.

To further evaluate the proposed framework beyond the distillation setting, we compare it with representative methods from other privacy paradigms, including PATE, DP-SGD, HierarchicalDP, and QL-PGD. The results are summarized in Table 4. Unlike Table 3, which compares against heterogeneous distillation baselines, Table 4 is intended as a cross-paradigm comparison. In particular, these methods are based on different privacy mechanisms, such as teacher aggregation, gradient perturbation, feature-level perturbation, or regularization, whereas our framework protects distilled samples and teacher logits through secure sharing and performs collaborative distillation on the reconstructed knowledge.

It should also be noted that the original papers of these baselines do not report the privacy metric used in this work. Therefore, for all methods in Table 4, we re-evaluate privacy under the same black-box membership inference attack protocol described in Section 4.1.4, and report the corresponding privacy accuracy as Priv Acc (Ab). This unified evaluation makes the comparison more consistent under our threat model.

Several observations can be drawn from Table 4. First, on CIFAR-10, DP-SGD with a large privacy budget achieves the strongest utility among the non-ours baselines, reaching 56.59% accuracy, while the stricter DP-SGD setting with a smaller privacy budget drops to 53.85%. PATE and HierarchicalDP perform substantially worse, with test accuracies of 40.59% and 48.56%, respectively. QL-PGD achieves 60.99% accuracy, which is the strongest non-DP baseline in this group. In comparison, our method reaches 64.36% accuracy while keeping Priv Acc (Ab) at 49.62%, which is closer to random guessing than QL-PGD and competitive with the DP-based methods. This indicates that the proposed framework can preserve stronger utility than these representative privacy baselines while still maintaining low membership leakage.

Second, on STL-10, the proposed method achieves the highest utility among all compared methods, reaching 49.51% accuracy. In contrast, PATE drops sharply to 26.07%, HierarchicalDP reaches 34.75%, and QL-PGD reaches 44.01%. The two DP-SGD settings obtain 40.61% and 42.11%, respectively. In terms of privacy, most DP-based methods keep Priv Acc (Ab) near 50%, which is expected from noise-based or aggregation-based privacy mechanisms. Our method also keeps privacy accuracy near random guessing at 49.85%, while achieving clearly better utility than all of these alternatives. This result suggests that the proposed secure distillation framework can remain

competitive not only against distillation-based baselines, but also against representative methods from the differential privacy and regularization paradigms.

Overall, Table 4 shows that the proposed framework provides a favorable privacy–utility trade-off across paradigms. Compared with representative DP-based methods, our approach avoids the large utility degradation often caused by strong perturbation. Compared with regularization-based privacy defenses, it better controls membership leakage while maintaining competitive accuracy. These results further support the effectiveness of combining secure sharing and knowledge alignment in collaborative distillation.

Table 4. Comparison with differential privacy and regularization baselines.

Dataset	Method	Category	Test Acc (%)	Priv Acc (Ab) ↓
CIFAR-10	PATE	DP-based	40.59	50.01
	DP-SGD (ε=198.5)	DP-based	56.59	50.49
	DP-SGD (ε=50.2)	DP-based	53.85	50.14
	HierarchicalDP	Feature-DP	48.56	50.99
	QL-PGD	Regularization	60.99	50.34
	Ours (distilled 500 images)	Secure Distillation	64.36	49.62
STL-10	PATE	DP-based	26.07	50.62
	DP-SGD (ε=198.5)	DP-based	42.11	49.42
	DP-SGD (ε=50.2)	DP-based	40.61	49.72
	HierarchicalDP	Feature-DP	34.75	49.58
	QL-PGD	Regularization	44.01	54.44
	Ours (distilled 500 images)	Secure Distillation	49.51	49.85

4.5 Effect of Teacher Quality Imbalance

A central motivation of the proposed framework is to handle inconsistency under unequal teacher quality. To isolate this factor, we study the effect of different teacher combinations under the MPC-protected training setting, i.e., all results in this subsection are obtained after secure sharing has already been introduced. The results are summarized in Table 5.

On CIFAR-10, the single-teacher secure baseline using the 95% teacher reaches 65.61% accuracy. When two strong teachers are combined, the result becomes 65.06%, suggesting that simply increasing the number of strong teachers does not necessarily improve the student. When a weaker 80% teacher is added without KA, the accuracy drops to 63.94%, confirming that low-quality supervision can harm collaborative distillation even when the entire pipeline is protected by MPC. After enabling KA, the performance recovers to 64.36%, indicating that the alignment mechanism successfully suppresses the negative influence of the weaker teacher.

A similar trend is observed on STL-10. Under the same MPC-protected setting, the single-teacher baseline using the 85% teacher achieves 41.88% accuracy, while pairing an 85% teacher with a weaker 70% teacher without KA reduces the accuracy to 41.83%. After applying KA, the performance improves substantially to 49.51%. These observations confirm that teacher-quality imbalance remains a genuine challenge even after secure protection is introduced, and that the proposed alignment mechanism is particularly valuable in mitigating inconsistent supervision under the MPC setting.

Table 5. Effect of teacher quality imbalance under the MPC setting.

Teacher A	Teacher B	Dataset	Test Acc (%)	Priv Acc (Ab)
95%	none	CIFAR-10	65.61	50.22
95%	95%	CIFAR-10	65.06	51.83
95%	80% (w/o KA)	CIFAR-10	63.94	50.09
95%	80% (w/ KA)	CIFAR-10	64.36	49.62
85%	none	STL-10	41.88	49.25
85%	85%	STL-10	43.61	48.39
85%	70% (w/o KA)	STL-10	41.83	49.68
85%	70% (w/ KA)	STL-10	49.51	49.85

4.6 Cross-Domain Generalization under Heterogeneous Data

To further assess robustness under heterogeneous data distributions, we conduct cross-domain experiments in which the two institutions use different datasets. In this experiment, the CIFAR-10 side uses the teacher pair with accuracies 95.36% / 80.61%, while the STL-10 side uses the teacher pair with accuracies 84.95% / 71.08%. The results are shown in Table 6.

When both institutions use CIFAR-10, our method reaches 64.36% accuracy. When both institutions use STL-10, it reaches 49.51% under the corresponding MPC + KA configuration. More importantly, when one institution uses CIFAR-10 and the other uses STL-10, the proposed framework still achieves 49.47% test accuracy while keeping privacy accuracy near random guessing. This result is important because it shows that the framework is not limited to homogeneous collaborative settings. Even when institutions differ substantially in data distribution, aligned teacher supervision remains sufficiently informative to train a meaningful student model.

Table 6. Cross-domain generalization under heterogeneous data.

Dataset A	Dataset B	Method	Test Acc (%)	Priv Acc (Ab)
CIFAR-10	CIFAR-10	MPC + KA	64.36	49.62
CIFAR-10	STL-10	MPC + KA	49.47	50.05
STL-10	STL-10	MPC + KA	49.51	49.85

4.7 Ablation Study

Finally, we study the contribution of Knowledge Alignment after secure protection has already been introduced. To keep the ablation focused, we compare the MPC-protected multi-teacher setting without KA against the full MPC + KA model. The results are summarized in Table 7.

On CIFAR-10, adding KA on top of MPC-protected multi-teacher training improves the accuracy from 63.94% to 64.36%, corresponding to a +0.42% gain. On STL-10, the improvement is much larger, increasing from 41.88% to 49.51%, i.e., +7.63%. In both cases, privacy accuracy remains close to 50%, indicating that the alignment mechanism improves utility without compromising privacy protection. The larger gain on STL-10 further suggests that KA becomes more beneficial as the task becomes harder and the supervision becomes less stable. Since the complete single-teacher baselines are already reported in Table 2, we keep the ablation table focused on the incremental effect of KA under the MPC setting, rather than repeating baseline rows.

Table 7. Ablation study: Effect of Knowledge Alignment under MPC setting.

Configuration	Dataset	Test Acc (%)	Priv Acc (Ab)
MPC (no KA)	CIFAR-10	63.94	50.09
MPC + KA	CIFAR-10	64.36	49.62
MPC (no KA)	STL-10	41.88	49.25
MPC + KA	STL-10	49.51	49.85

4.8 Qualitative Visualization

In addition to quantitative evaluation, we provide qualitative visualizations to illustrate the role of each key component in the framework. The additive secret shares observed by different parties are shown in Fig. 4. These shares appear visually indistinguishable from random noise, providing intuitive evidence that a single transmitted share does not reveal the original image content. Finally, the reconstructed samples are shown in Fig. 5, demonstrating that the secure sharing mechanism preserves sufficient fidelity for downstream student training once reconstruction is performed inside the secure workflow. Together with the distilled samples shown earlier in Fig.3, these visualizations complement the quantitative privacy results by showing both the hiding effect of secure sharing and the recoverability required for effective collaborative distillation.

Fig. 4. Visualization of additive secret shares observed by different parties

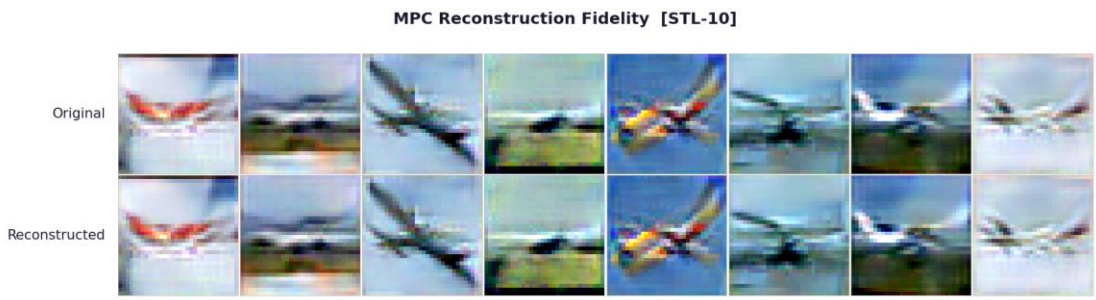

Fig. 5. Visualization of reconstructed images after secure recovery

4.9 Discussion

The results suggest that the effectiveness of the proposed framework comes from the complementary roles of secure sharing and knowledge alignment. The MPC layer mainly protects the exchange of distilled images and teacher logits, rather than the entire optimization process. This

design helps explain why the framework can substantially reduce intermediate privacy leakage while introducing only limited performance degradation.

Another important observation is that heterogeneous supervision is useful only when it is made comparable. When teachers are trained on data with different quality levels or distributions, their logits may differ in confidence, calibration, and semantic tendency. Directly reusing such signals can therefore destabilize student training. The role of KA is to reduce this mismatch, which also explains why its benefit is much larger on STL-10 and under teacher-quality imbalance.

At the same time, the current framework does not completely remove the difficulty of heterogeneous collaboration. The remaining gap in cross-domain settings suggests that secure sharing and lightweight alignment make collaboration feasible, but not fully optimal. This indicates that future improvements should focus on stronger teacher reliability modeling and more adaptive alignment under severe heterogeneity.

5. Conclusion

In this paper, we studied privacy-preserving multi-teacher knowledge distillation in heterogeneous collaborative learning. Existing distillation-based privacy-preserving methods remain limited in three important respects: distilled knowledge may still leak sensitive information, the exchange process is often insufficiently protected, and heterogeneous teachers may provide inconsistent supervision. To address these issues, we proposed a unified framework that combines MPC-based secure sharing with Knowledge Alignment.

The proposed method protects distilled images and teacher logits during exchange, while aligning heterogeneous teacher outputs before student training. In this way, it jointly addresses two coupled challenges in collaborative distillation: secure knowledge sharing and reliable multi-teacher fusion.

Experiments on CIFAR-10 and STL-10 demonstrated that the framework achieves a favorable privacy–utility trade-off, with Knowledge Alignment providing up to 7.63% accuracy gains under heterogeneous settings.Cross-domain experiments further demonstrated that the framework retains meaningful collaborative generalization even when institutions operate on different datasets.

Although effective, the current framework still has limitations. The present design mainly protects exchanged distilled knowledge rather than performing fully encrypted end-to-end optimization, and the experiments are limited to image classification benchmarks with a small number of collaborating institutions. Future work may extend the framework to larger-scale multi-institution settings, more complex data modalities, and more adaptive alignment strategies under severe heterogeneity.

Overall, this work shows that practical privacy-preserving collaborative distillation requires not only shielding raw data, but also protecting exchanged intermediate knowledge and regulating the compatibility of heterogeneous supervision.

Reference List:

[1] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[2] G. Litjens *et al*., "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.

[3] A. Brauneck, B. van der Sloot, M. M. A. de Reuver, and M. Pechenizkiy, "Federated machine learning, privacy-enhancing technologies, and data protection law in medical research: Scoping review," *JMIR Medical Informatics*, vol. 11, e41588, 2023.

[4] S. Z. El Mestari, G. Lenzini, and H. Demirci, "Preserving data privacy in machine learning systems," *Computers & Security*, vol. 137, 103605, 2024.

[5] A. Guerra-Manzanares, L. J. Lechuga Lopez, M. Maniatakos, and F. E. Shamout, "Privacy-preserving machine learning for healthcare: Open challenges and future perspectives," in *Distributed Computing to Support the Future Internet*, Cham, Switzerland: Springer, 2023.

[6] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *Proc. IEEE Symposium on Security and Privacy (SP)*, 2017, pp. 3–18.

[7] M. Nasr, R. Shokri, and A. Houmansadr, "Comprehensive privacy analysis of deep learning: Passive and active white-box inference attacks against centralized and federated learning," in *Proc. IEEE Symposium on Security and Privacy (SP)*, 2019, pp. 739–753.

[8] N. Carlini, C. Liu, Ú. Erlingsson, J. Kos, and D. Song, "The secret sharer: Evaluating and testing unintended memorization in neural networks," in *Proc. USENIX Security Symposium*, 2019, pp. 267–284.

[9] N. Carlini *et al*., "The privacy onion effect: Memorization is relative," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[10] M. Abadi *et al*., "Deep learning with differential privacy," in *Proc. ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2016, pp. 308–318.

[11] N. Papernot, S. Song, I. Mironov, A. Raghunathan, K. Talwar, and Ú. Erlingsson, "Scalable private learning with PATE," in *Proc. International Conference on Learning Representations (ICLR)*, 2018.

[12] T. D. Pham, B. D. Nguyen, S. T. Mai, and V. C. Ta, "QL-PGD: An efficient defense against membership inference attack," *Journal of Information Security and Applications*, vol. 92, 104095, 2025.

[13] G. Liu, W. Zhang, X. Wang, S. King, and S. Yu, "A membership inference and adversarial attack defense framework for network traffic classifiers," *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 2, pp. 317–332, 2025.

[14] M. Nasr, R. Shokri, and A. Houmansadr, "Machine learning with membership privacy using adversarial regularization," in *Proc. ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2018, pp. 634–646.

[15] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.

[16] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *International Journal of Computer Vision*, vol. 129, pp. 1789–1819, 2021.

[17] Y. Zhang, T. Xiang, T. M. Hospedales, and H. Lu, "Deep mutual learning," in *Proc. IEEE/CVF*

Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 4320–4328.
[18] V. Shejwalkar and A. Houmansadr, "Membership privacy for machine learning models through knowledge transfer," in *Proc. AAAI Conference on Artificial Intelligence (AAAI)*, 2021, pp. 9549–9557.
[19] F. Faisal, C. K. Leung, N. Mohammed, and Y. Wang, "Privacy-preserving learning via data and knowledge distillation," in *Proc. IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 2023.
[20] T. Wang, J.-Y. Zhu, A. Torralba, and A. A. Efros, "Dataset distillation," *arXiv preprint arXiv:1811.10959*, 2018.
[21] G. Cazenavette, T. Wang, A. Torralba, A. A. Efros, and J.-Y. Zhu, "Dataset distillation by matching training trajectories," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
[22] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.
[23] P. Kairouz *et al*., "Advances and open problems in federated learning," *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.
[24] C. Wu, F. Wu, L. Lyu, Y. Huang, and X. Xie, "Communication-efficient federated learning via knowledge distillation," *Nature Communications*, vol. 13, 2032, 2022.
[25] J. Shao, F. Wu, and J. Zhang, "Selective knowledge sharing for privacy-preserving federated distillation without a good teacher," *Nature Communications*, vol. 15, 349, 2024.
[26] D. Li and J. Wang, "FedMD: Heterogeneous federated learning via model distillation," *arXiv preprint arXiv:1910.03581*, 2019.
[27] S. Zheng, J. Hu, G. Min, and K. Li, "Mutual knowledge distillation-based personalized federated learning for smart edge computing," *IEEE Transactions on Consumer Electronics*, vol. 71, no. 2, 2025.
[28] Y. Chen, J. Zhang, Z. Wang, W. Dai, H. Gao, and S. Deng, "Privacy-preserving knowledge distillation in latency-critical federated task offloading for consumer IoT networks," *IEEE Transactions on Consumer Electronics*, vol. 71, no. 1, pp. 239–251, 2025.
[29] S. You, C. Xu, C. Xu, and D. Tao, "Learning from multiple teacher networks," in *Proc. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2017, pp. 1285–1294.
[30] U. Milasheuski, L. Barbieri, B. Camajori Tedeschini, M. Nicoli, and S. Savazzi, "On the impact of data heterogeneity in federated learning environments with application to healthcare networks," *arXiv preprint arXiv:2404.18519*, 2024.
[31] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-IID data," *arXiv preprint arXiv:1806.00582*, 2018.
[32] Xidian University NSS Lab, "NssMPClib: Secure Multi-Party Computation Library," software, 2024. Available: https://github.com/XidianNSS/NssMPClib